

CURRICULUM VITAE

GUOXIA ZHAO

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OBJECTIVES

• Process Chemist/ Entry Level Scientist in Chemical or Pharmaceutical/Biotech Companies or Related Fields Including Petrochemical, Chemical Engineering Industries.

PROFILE OF QUALIFICATIONS and EXPERTISE

- Over nine years of research experience in organic/synthetic/medicinal chemistry, nucleoside anti-viral drugs, anti-tumor drug design and total synthesis, nucleoside and nucleotide chemistry, heterocyclic chemistry and natural product synthesis.
- High skills in bioactive small molecules preparation and characterization.
- Extensive experience in natural products purification, structural elucidation, and lead optimization.
- Solid experience in large scale active pharmaceutical ingredient (API) synthesis
- Knowledge and Hands-on experience in modern chromatographic techniques (GC, normal and reverse phase HPLC), Mass spectroscopy, FT-IR, UV, GC-MS and ^1H and ^{13}C NMR (1D & 2D) including HMQC, HMBC and NOESY.
- Some experience in fluorescence (steady-state and time-resolved), AFM, TEM, HRTEM, Low-angle Laser Light Scattering (LALLS), and Dynamic Light Scattering (DLS).
- Excellent computer skills in Chem Office including ChemDraw, ChemSketch, and 1D/2D NMR Processor, EndNote, and Microsoft Office Suite software.
- Proficient in Web of Science, Scopus, PubMed, and SciFinder search engines.
- Over three years of teaching experience in general and organic chemistry classes, and their laboratories for undergraduate students.
- Well-organized multitasker with strong detail orientation.
- Self-driven team player with strengths in communications and developing interpersonal relationships, good people skill and team work.
- Possesses excellent planning, organizing, time management and decision-making skills.
- Strategically coordinate and collaborate with various professionals to maximize performance in facilitating goals and attaining operational excellence.

OTHER PROFESSIONAL EXPERIENCE

• Sep. 2010 – Aug. 2011, Research Assistant

Department of Chemistry and Biochemistry, Old Dominion University, Norfolk, VA

Working in Professor Lesley H. Greene's lab on protein biochemistry, folding, and NMR

• June 2009 – Aug. 2010, Adjunct Assistant Professor

Department of Chemistry and Biochemistry, Old Dominion University, Norfolk, VA,
Teaching Organic Chemistry Laboratory for undergraduate students

EDUCATION

- Aug. 2004 – May 2009

Ph. D. in Chemistry specialized in Organic and Medicinal Chemistry, Auburn University,
Auburn, AL, USA

Advisor: Professor Stewart W. Schneller

- Aug. 2001 – July 2004

M. S. in Chemistry specialized in Organic Chemistry, University of Science & Technology of China, Hefei,
P. R. China

Advisor: Professor Guan-wu Wang and Professor Lifeng Yan

- Aug. 1997 – July 2001

B. S. in Chemistry, Qufu Normal University, Qufu, Shandong, P. R. China

ACADEMIC AWARDS AND HONORS

- June 2005 – May 2009 Graduate Student Research Scholarship of Department of Chemistry and Biochemistry of Auburn University

- Sep. 2003 – July 2004 Excellent Graduate Scholarship in School of Chemistry and Material Science of USTC

- Sep. 1998 – July 2001 Excellent Student Scholarship and Excellent Graduate Fellowship of Department of Chemistry of Qufu Normal University

RESEARCH EXPERIENCES

- Aug. 2004 – May 2009

Projects: • Designed and synthesized carbocyclic sinefungin, which is the structural analog of S-adenosylmethionine (AdoMet), and further characterized its biological activities.

- Designed and synthesized the analog of carbocyclic sinefungin, which is a potential inhibitor of AdoMet methyltransferase, and established two stereocenters by efficient asymmetric synthetic and chiral auxiliary approaches.

- Designed and synthesized the analog of sinefungin with efficiently enantioselective approaches.

- Aug. 2001 – July 2004

Project: • Synthesized a series of novel C₆₀ derivatives, investigated their physical activities by UV-Vis, FT-IR, fluorescence (steady-state and time-resolved), and characterized with AFM, TEM and HRTEM; and also studied their aggregation status (monomer, dimer or trimer) in different conditions by LALLS) and DLS.

TEACHING EXPERIENCES

- Aug. 2004 – May 2005 Graduate Teaching Assistant, Auburn University, Auburn, AL

Course: General and Organic Chemistry Laboratories

- Aug. 2001 – July 2002 Graduate Teaching Assistant, Department of Chemistry, University of Science & Technology of China, Hefei, P.R. China

SERVICE ON EDITORIAL BOARDS

Jan. 2010 – present, review manuscripts for the following journals: *Chemical Research in Chinese Universities*, *New Journal of Chemistry*, *Journal of Materials Chemistry*

PUBLICATIONS

- 1) **G. Zhao**, X. Yin and S.W. Schneller*, Formal total synthesis the analog of carbocyclic sinefungin, *manuscript in preparation*.
- 2) **G. Zhao**, X. Yin and S. W. Schneller*, Total synthesis the analog of sinefungin, *manuscript in preparation*.
- 3) L.H. Greene*§, H. Li§, J. Zhong, **G. Zhao** and K. Wilson, Folding of an all-helical Greek-key protein monitored by quenched-flow hydrogen-deuterium exchange and NMR spectroscopy. *Eur. Biophys. J.*, 2012, 41, 41-51. (§ L.H. Greene and H. Li, co-first authors)
- 4) X. Yin, **G. Zhao** and S. W. Schneller*, Carbocyclic sinefungin, *Tetrahedron Lett.*, 2007, 48, 4809-4811.
- 5) G.W. Wang*, **G. Zhao** and L. Yan, Investigation on the aggregation properties of cationic [60]fullerene derivative, *Chin. Sci. Bull.*, 2004, 49, 1441-1445.

CONFERENCE ABSTRACTS

- 1) **G. Zhao**, X. Yin and S.W. Schneller*, Synthesis of the analogue of carbocyclic sinefungin, 235th ACS National Meeting, April 6-10, 2008, New Orleans, LA, United States.
- 2) X. Yin, **G. Zhao** and S.W. Schneller*, Total synthesis of carbocyclic sinefungin, 233rd ACS National Meeting, March 25-29, 2007, Chicago, IL, United States.

REFERENCES

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